MoonTrack

Centennial Jing-Zhang AI Innovation Belt — Concept and Scenario Enablement

A3 booklet — flagship depth: agent.3, Xiaoyue River Scenario Enablement Wing

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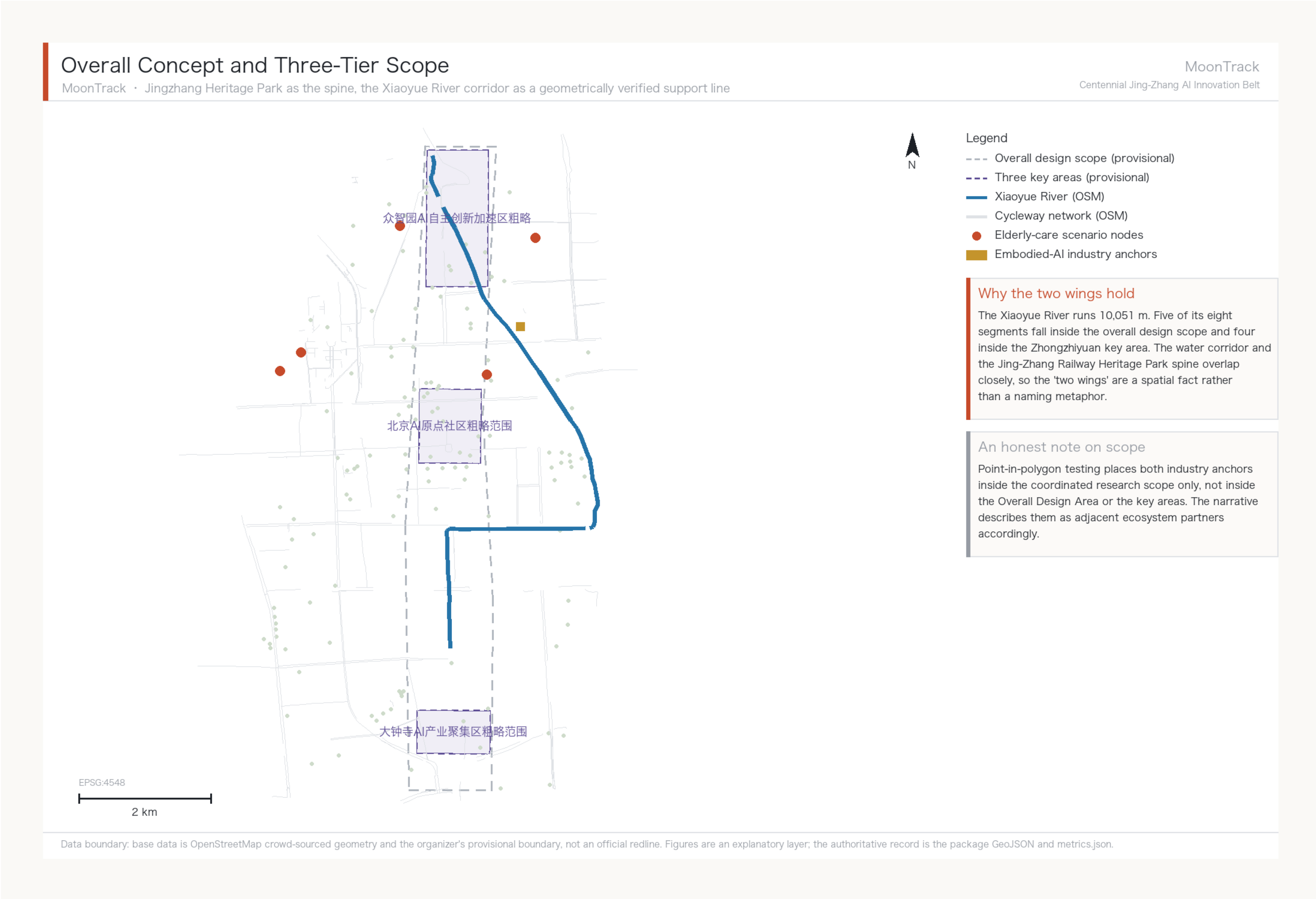
All figures are derived from the package GeoJSON and metrics.json and are reproducible.

Overall Concept and Three-Tier Scope

A relay of self-reliant innovation



Spatial basis for the two wings

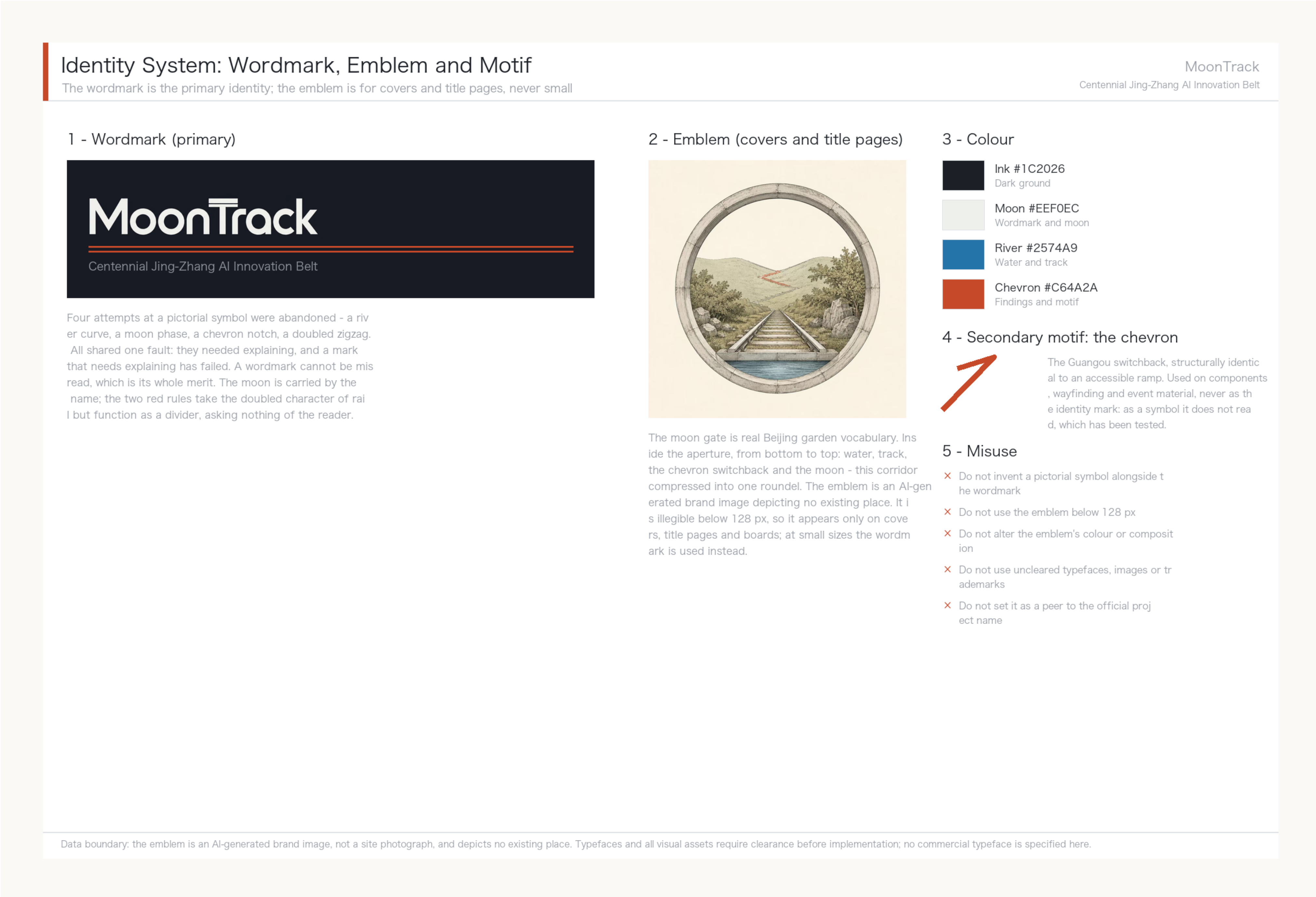
The Xiaoyue River runs 10,051 m; five of its eight segments fall inside the design scope and four inside the Zhongzhiyuan key area. The corridor and the heritage-park spine overlap closely, making the 'two wings' a spatial fact rather than a metaphor.

Scope attribution

Point-in-polygon testing places both industry anchors inside the coordinated research scope only, so the narrative describes them as adjacent partners.

Identity System: Wordmark, Emblem and Motif

The wordmark is primary; the emblem is for covers and title pages, never small



A mark that needs explaining has failed

Four pictorial symbols were tried - a river curve, a moon phase, a chevron notch, a doubled zigzag - and all shared one fault: viewers could not read them. The pictorial symbol was therefore abandoned for a wordmark, which cannot be misread. The moon is carried by the name; the two red rules are a divider and ask nothing of the reader.

The emblem: a moon gate

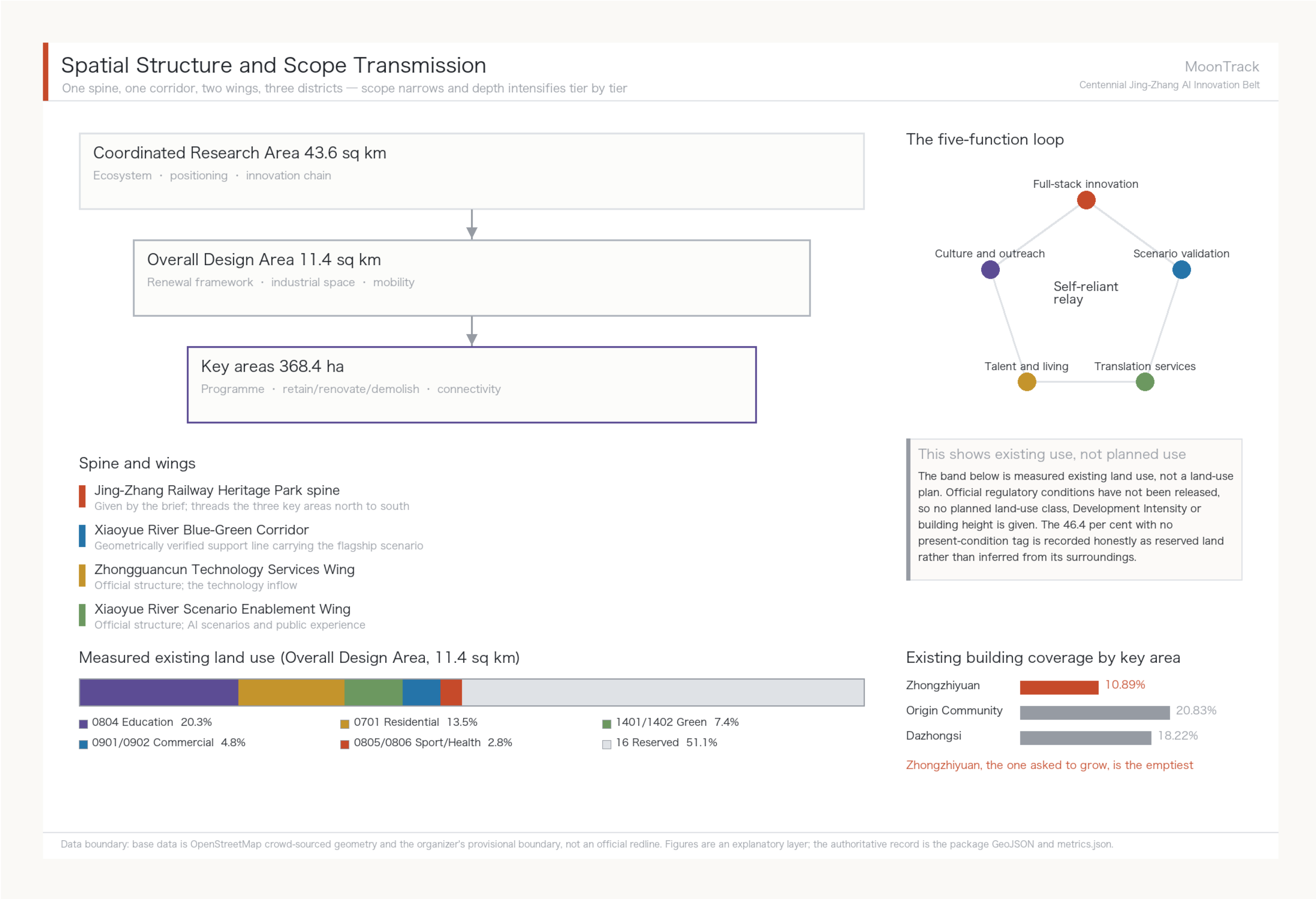
The moon gate is real Beijing garden vocabulary. Inside the aperture, from bottom to top: water, track, the chevron switchback and the moon - this corridor compressed into one roundel. It is illegible below 128 px, so it appears only on covers, title pages and boards.

Authorship

The Latin logotype and the emblem are both AI-generated and used exactly as generated; the Chinese characters, rules and layout are set by make_identity.py. Both are declared in report/copyright_statement.md. No commercial typeface is specified.

Spatial Structure and Transmission

Scope narrows, depth intensifies

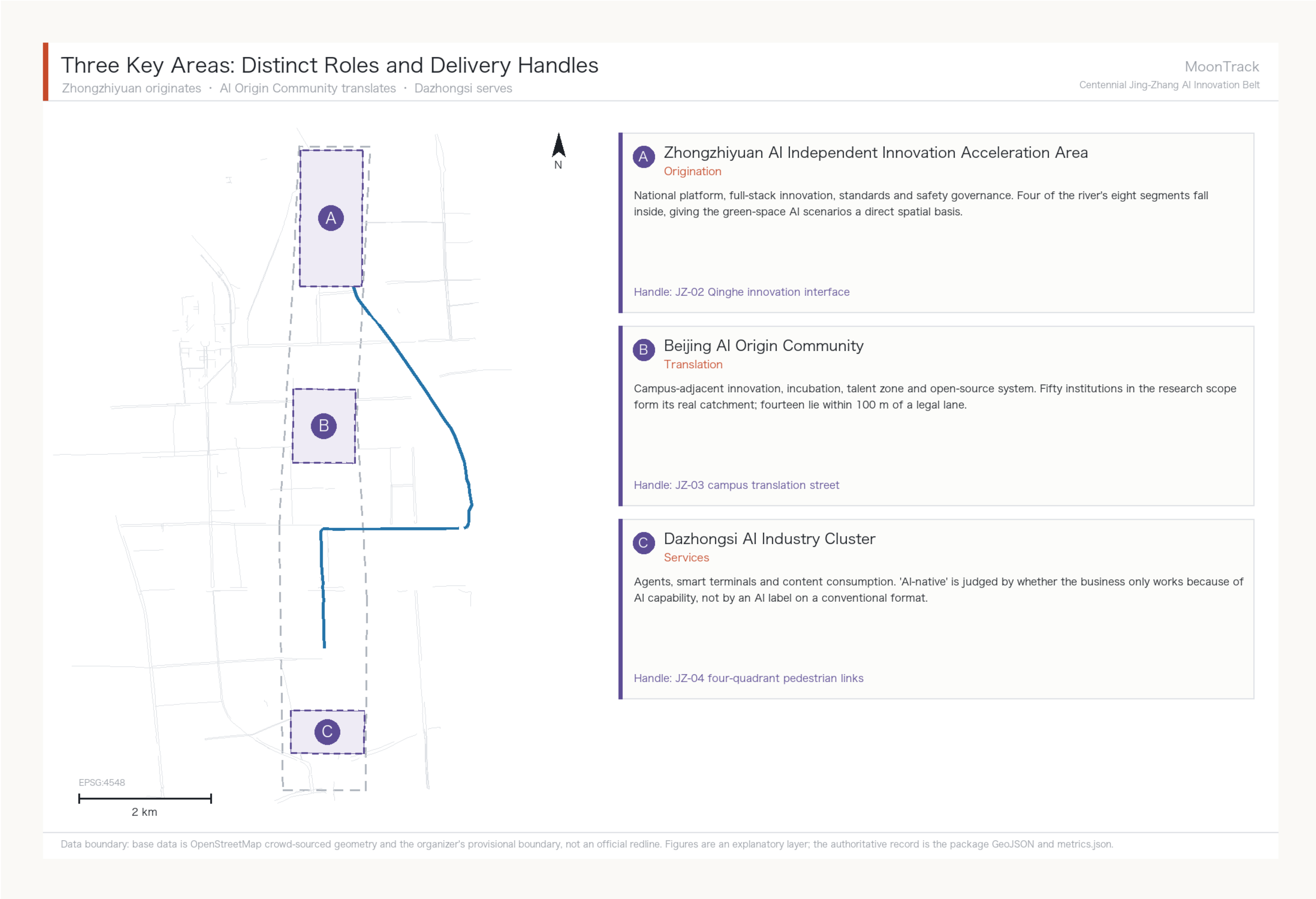


Why not a land-use colour plan

Official regulatory conditions have not been released, so land-use classes, intensity and heights cannot properly be drawn. This tier gives structure and transmission logic instead of falsely precise colour blocks; design_depth_matrix.json records the true depth of each item.

Three Key Areas

Originate / translate / serve



Distinct roles

Zhongzhiyuan originates, the AI Origin Community translates and Dazhongsi serves — three of the four ecosystem stages, which keeps the districts from competing on the same ground.

Key-area boundary status

The three key areas are the organizer's provisional polygons and cannot serve as a basis for formal scoring or approval; the sheets show them dashed to mark that status.

Measured Cycleway Connectivity

The proposal's strongest transport finding

Core finding: 0 of 5

Network-reachable elderly facilities: zero of five. Four sit in different components from the candidate depot; the fifth is connected but its final 652 m has no legal right of way. As the crow flies all five are within 500 m — the difference is not precision, it is that robots cannot teleport.

56 m unlocks 30 km

The three shortest gaps measure 9 m, 12 m and 35 m. If real and passable, the main network grows from 27,560 m to 57,770 m. This sets the phasing: verify the breaks before buying robots.

What it cannot prove

Topological connectivity is not passability. Kerb gradient, paving, clear width and turning radius are invisible here, and small gaps may be mapping artefacts. JZ-07 is therefore a field-verification list, not a construction brief.

Metric Evidence Chain

All 50 known metrics are cited in the narrative

Metric Evidence Chain and Confidence Tiers

All 50 known metrics are cited in the narrative; the line between what can and cannot be computed matters more than the numbers

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Tier 1 · Direct from geometry
GeoJSON → area, length, count

Metric	Value	Confidence
site_area_sqm	11,412,825 sqm	medium
key_area_count	3	high
xiaoyuehe_total_length_m	10,051 m	medium
cycleway_total_length_m	142,150 m	medium
green_ratio	0.123	low
public_space_ratio	0.073	low

Tier 2 · Measured network reachability
The principal increment of this work

Metric	Value	Confidence
cycleway_network_connected_components	62	medium
cycleway_largest_component_share	0.149	medium
elderly_facilities_network_reachable	0 / 5	medium
elderly_facilities_within_100m_of_legal_lane	1 / 5	medium
robot_service_network_10min_m	5,430 m	low
gap_closure_unlocked_network_m	30,210 m	low

Tier 3 · Awaiting official controls
Deliberately left unknown

Metric	Value
floor_area_ratio	unknown
Height / density / setback	not stated
Road redline	not stated

Self-check status and evidence boundary

Once the four gates (structure, spatial, visual, professional evidence) pass, self_check_submission.py writes self_check.json and updates the manifest. Confidence labels are not rhetoric: green_ratio and public_space_ratio are marked low because the base data is crowd-sourced and cannot support a compliance conclusion; the gap_closure metrics are low because they are hypotheses awaiting field verification. floor_area_ratio is unknown with a stated reason — publishing that number before the official controls exist would be writing a guess as a control.

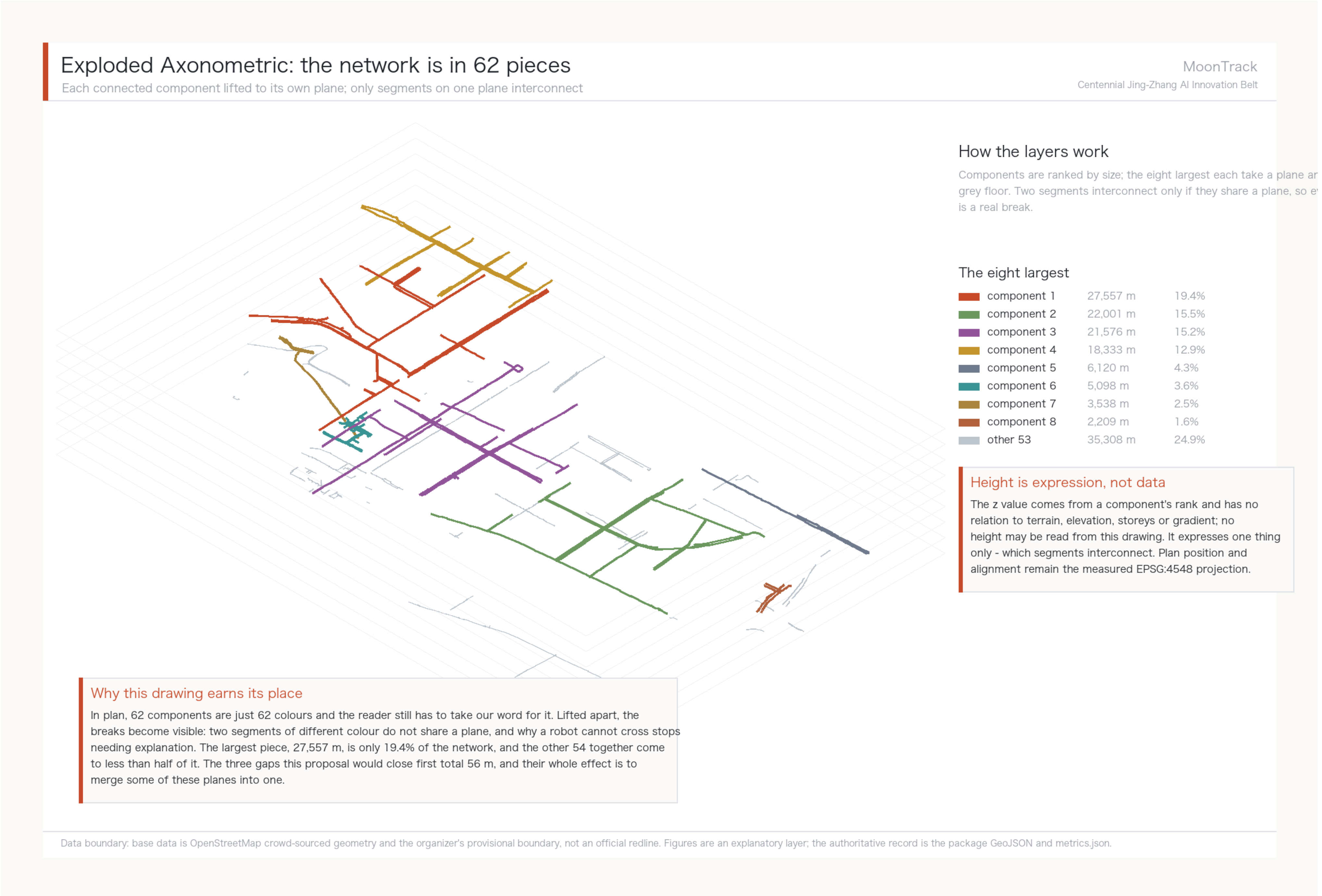
Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

The tiering is itself a finding

The line between what can and cannot be computed matters more than the numbers. Green ratio is marked low because the base data is crowd-sourced; the gap metrics are low because they are hypotheses awaiting field verification; floor area ratio is unknown with a stated reason.

Exploded Axonometric: the Network Is in 62 Pieces

Turning connectivity from a stated conclusion into something visible



A plan cannot show a break

In plan, 62 components are only 62 colours and the reader must take our word for it. Lifted apart, two segments of different colour visibly do not share a plane, and why a robot cannot cross stops needing explanation.

Even the largest piece is only 19.4%

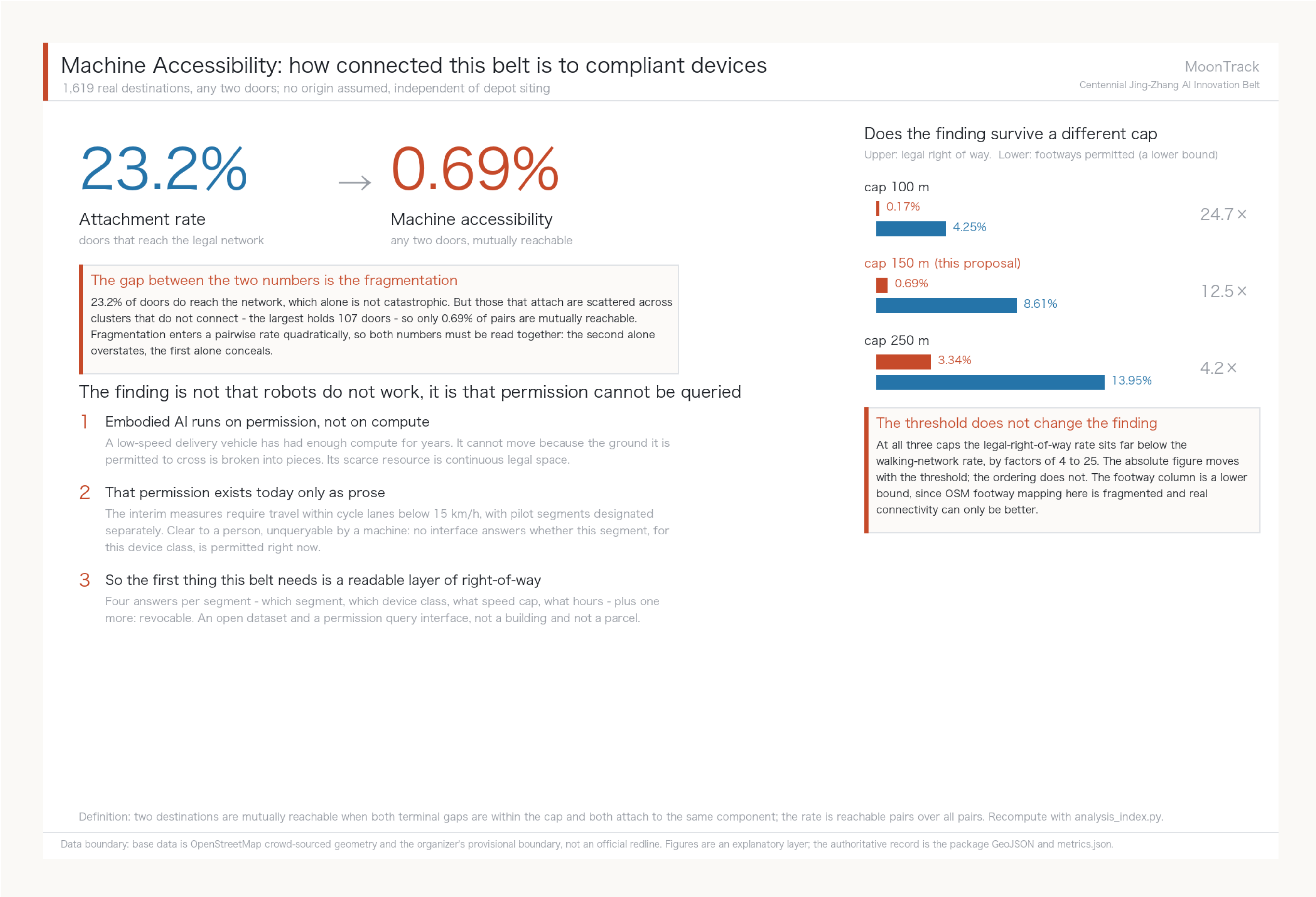
The largest component is 27,557 m, 19.4% of the network; the other 53 total 35,308 m, or 24.9%. The three gaps this proposal would close first come to 56 m, and their whole effect is to merge some of these planes into one.

Height is expression, not data

The z value follows a component's rank and has no relation to terrain, elevation, storeys or gradient; no height may be read from this drawing. Plan position and alignment remain the measured EPSG:4548 projection.

The Core Claim: Machine Accessibility and Readable Right-of-Way

A depot-independent measure that compresses the whole belt's fragmentation into one number



0.69% and 23.2% must be read together

Of 1,619 real destinations, 23.2% reach the legal network - but those that do are scattered across clusters that never connect, the largest holding 107, so only 0.69% of pairs are mutually reachable. The gap between the two numbers is the fragmentation itself.

No origin assumed, so moving the depot changes nothing

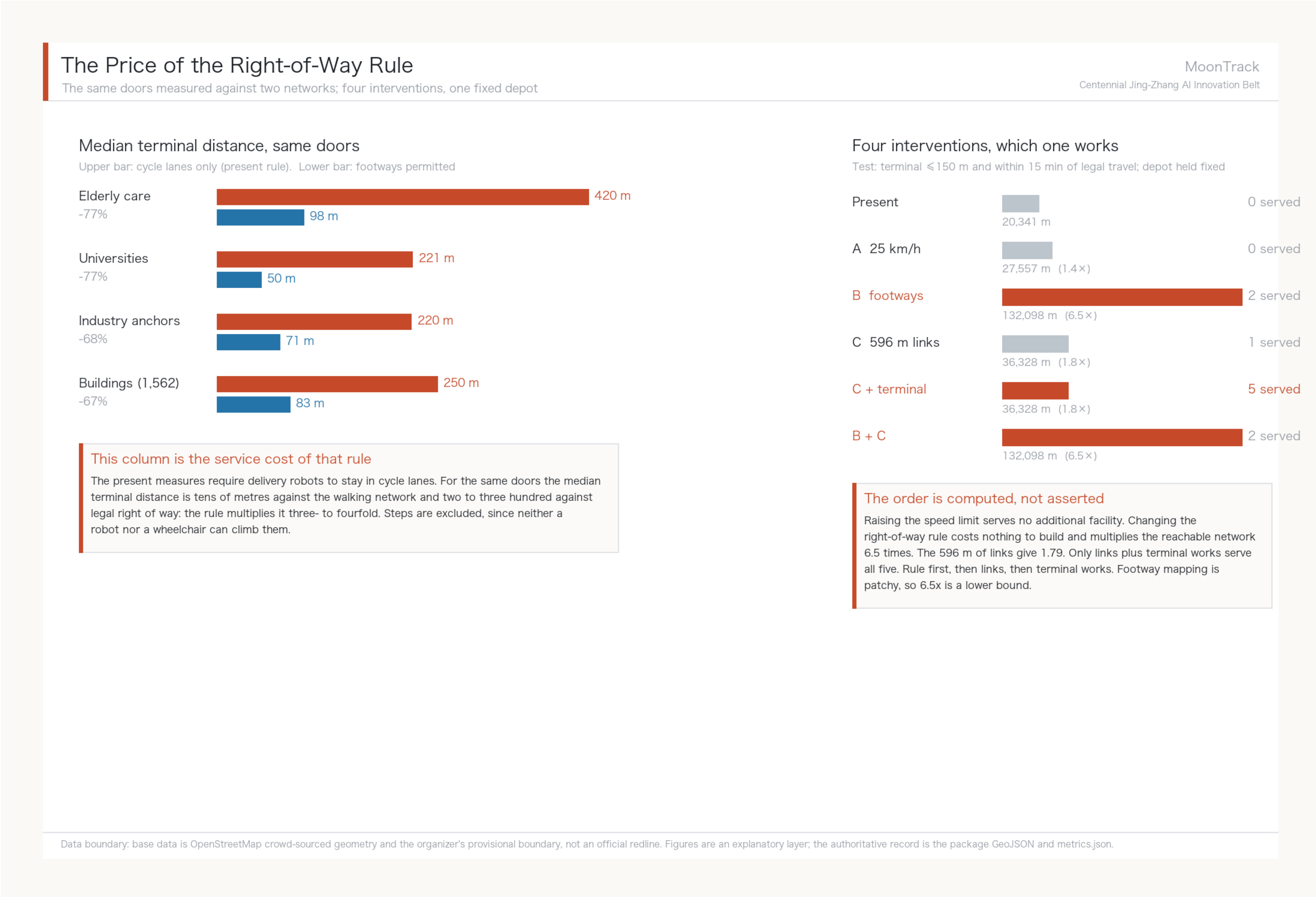
Every other conclusion here rests on a depot location. This measure assumes none: whether any two doors can reach each other does not depend on where a depot sits. Across three terminal caps the ratio runs 24.7x, 12.5x and 4.2x - the absolute figure moves, the ordering does not.

Hence: what this belt needs first is a readable layer of right-of-w

Embodied AI runs on permission, not on compute, and that permission exists today only as prose no machine can query. The proposal is four answers per segment - which segment, which device class, what speed cap, what hours - plus revocable. The same layer serves wheelchairs, walking frames and prams; the robot is simply what exposed the problem first.

The Cost of the Right-of-Way Rule

Four interventions compared on one basis



Speed does nothing

Raising the cap from 15 to 25 km/h serves not one additional facility and grows the 15-minute network by only 35%. Speed is not this corridor's constraint; the shape of the network is. Any recommendation built on relaxing the speed limit is void against this data.

The strongest lever costs nothing to build

Permitting low-speed devices on footways (steps excluded) grows the 15-minute network from 20,341 m to 132,098 m, a factor of 6.5. For the same doors a person on foot faces a median terminal gap of 82.8 m against the robot's 249.6 m. The robot fails not because the AI is not capable enough, but because it is only permitted on cycle lanes.

The sequence is computed, not asserted

Change the rule first (no construction, 6.5x), then close the links (596 m, 1.79x), then do the terminals (1,841 m, bringing in the last four). Only the terminal works make all five servable. The footway figures are a lower bound: OSM footway mapping is fragmented, so real connectivity can only be better.

Dispatch Ledger: 30 Tasks, 3 Successes

Every outcome decided by the measured graph, not synthetic trajectories

Round	Right-of-way state	Reachable	Delivered	Median gap	Worst gap
A	Cycle lanes as measured	2 / 10	0 / 10	230 m	652 m
B	**Plus proposed 596 m of links**	**6 / 10**	**1 / 10**	**230 m**	**652 m**
C	Counterfactual: footways permitted	2 / 10	2 / 10	76 m	188 m

Row B is the useful one

The 596 m of new links raises network reachability from 2 to 6 but deliveries only from 0 to 1, and the median terminal gap does not move at all. The construction lands, and most tasks are still blocked at the door rather than on the network - the same conclusion the connectivity analysis reached independently.

Audit completeness 30%, the ugliest number this package records about itself

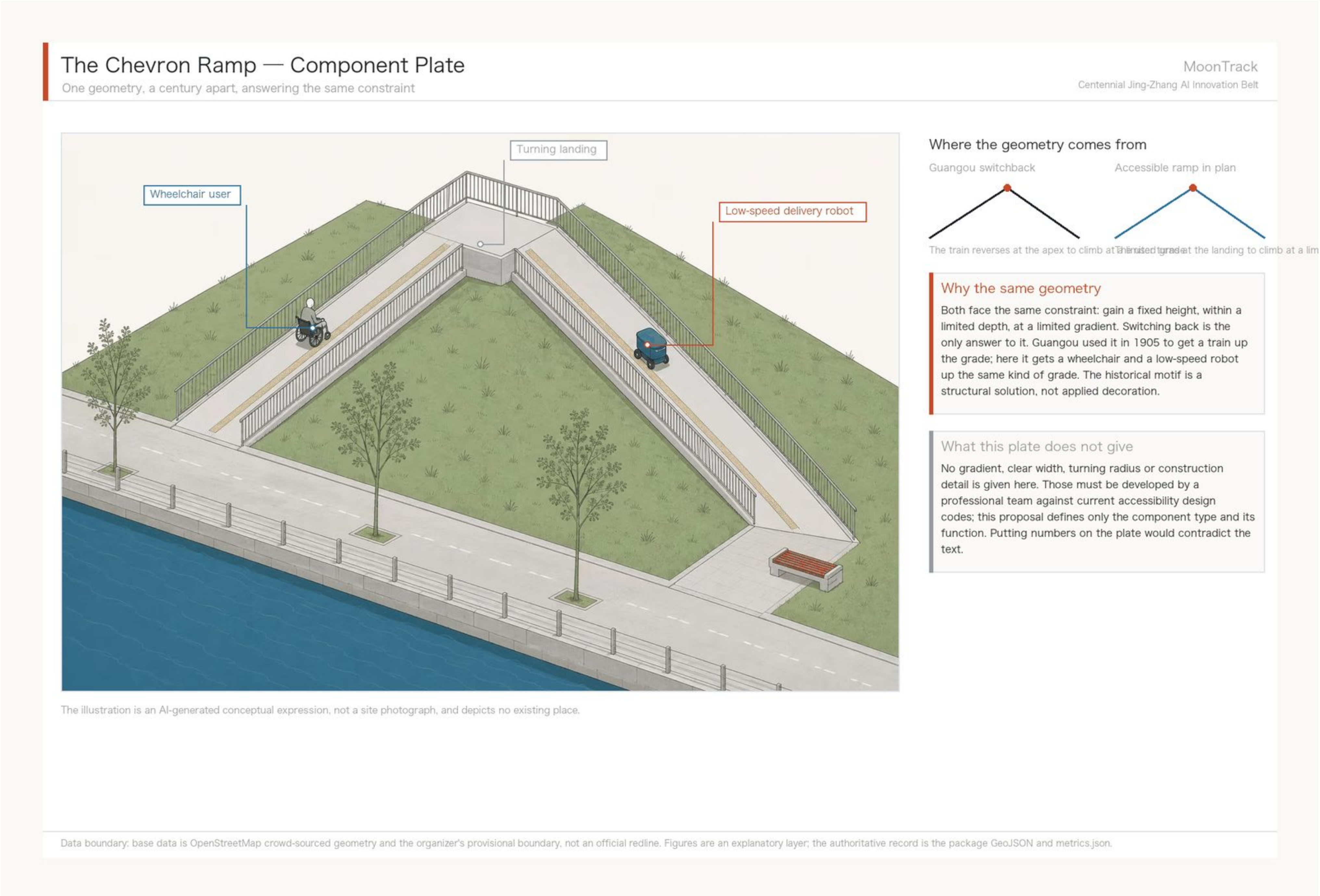
Two things drive it: rounds B and C depend on right-of-way that does not exist today, and for universities and industry anchors this proposal names no human handover party at all. The five elderly-care facilities have one, written into scenario card 02. The other five do not. That is a real operational gap, recorded in the ledger rather than left to hide behind the phrase 'the scenario is deliverable'.

No battery parameter enters the energy budget

Budget = straight-line distance x 1.5 x unit rate; consumption = legal route length x the same rate. The rate cancels, so '5 tasks over budget' is exactly the count of tasks whose legal route exceeds 1.5x the straight line - a purely geometric quantity. No claim is made about any vehicle's range.

Component 1: Chevron Ramp

The historical motif is a structural solution, not ornament



Why the same geometry

The Guangou switchback and an accessible ramp face one constraint: climbing a fixed rise at a limited gradient within a limited depth. Switching back is the only answer, which is why they take the same form.

Who it serves at once

Wheelchair users and low-speed delivery robots are bound by the same gradient limit, so one ramp answers both. If the robots leave, the ramp still serves wheelchairs, prams and luggage. This is where the reversibility principle lands.

What this sheet does not give

Gradient, clear width, turning radius and structure are not specified here; they belong to a professional team. The sheet carries only figures this proposal measured and rules its text already states. The illustration is an AI-generated concept image, not a photograph.

Component 2: Platform Unit

The phase-one deliverable; the handover boundary drives the form

The Platform Unit — Phase One Plate

Phase one is not all five connected: it is 82 m, one stop, one facility

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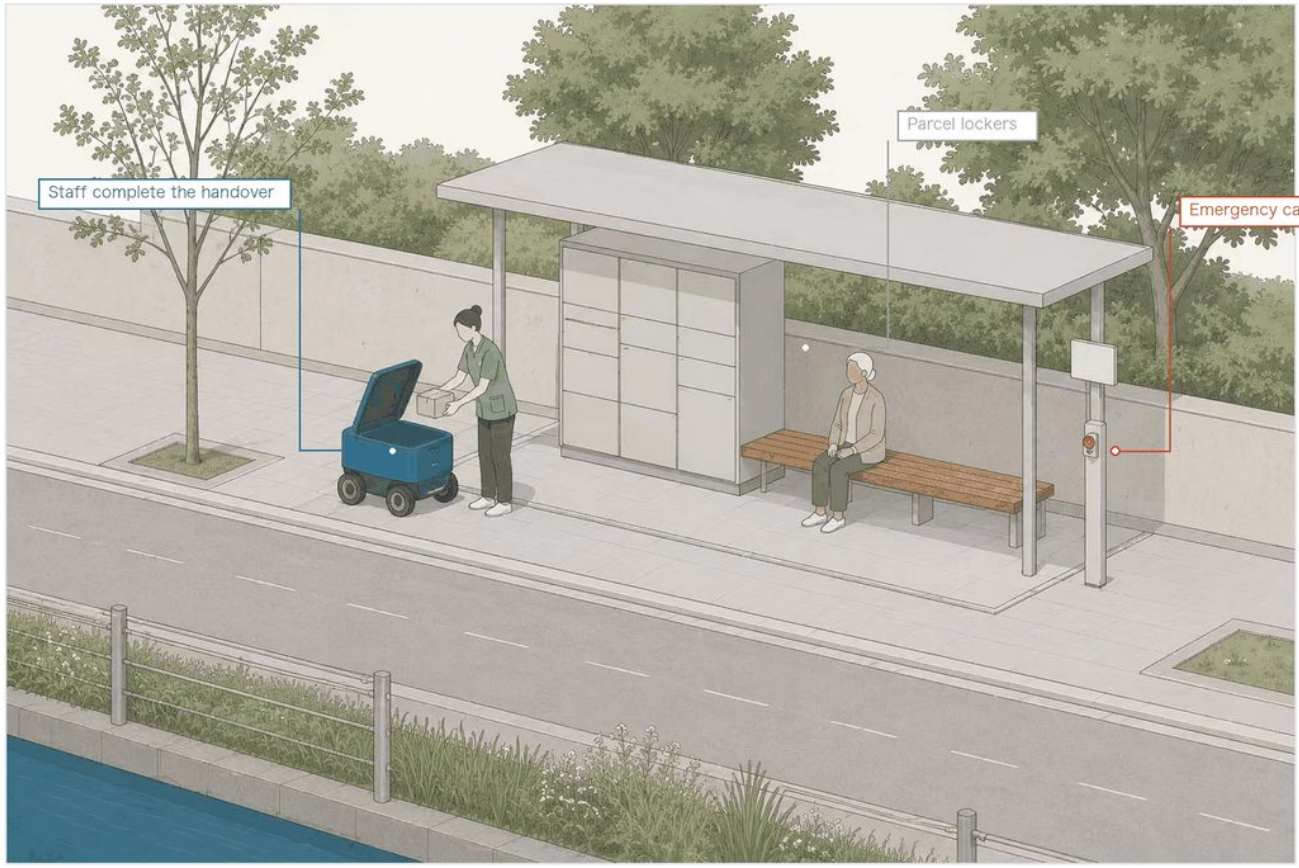


The illustration is an AI-generated conceptual expression, not a site photograph, and depicts no existing place.

This stop is a computed result

Phase-one link	82 m
Facilities served	1 / 5
Worst run once merged	14.0 min
Terminal access, five sites	1,950 m

The handover boundary is built into the component

The robot stops at the door and staff take delivery: it does not enter the building, does not interact with the resident directly, and makes no automated decision. Those three rules are not a line in an operations manual — they are why the unit is shaped this way, opening only to the street side with no opening on the building side.

Why phase one serves only one

Terminal distances for the five facilities are 109, 218, 420, 551 and 652 m. Drawing the line at 150 m, the distance at which a round trip to collect a delivery does not become a burden, only one qualifies. The remaining 1,841 m across four sites is walking and accessibility work, separately scoped and scheduled. Writing phase one as "all five connected" would look better and could not be built.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

Phase one serves one facility, and that is computed

Phase one: 82 m of link, one facility of five, 14.0 minutes to the furthest once joined. The other four have 1,841 m of terminal gap between them, past the 150 m threshold, so they belong to walking-and-cycling works rather than equipment procurement.

The boundary is not a note, it is the form driver

No entry indoors, no contact with the resident, no automated decisions. The cabinet therefore opens only to the street side and has no indoor-side aperture: staff take receipt on the street side, and the robot has no physical means of entering.

Reuse, not new build

All six robot scenario cards use existing facilities; none requires a dedicated new structure. The platform unit is a demountable module: when the pilot ends it comes out, leaving no permanent occupation of the footway.

Component 3: Observation Point

Where explainability stops being a principle and becomes a requirement

The Observation Point — Explainability Plate

Open to watch is not open data: oversight starts with being able to see what is happening

The illustration is an AI-generated conceptual expression, not a site photograph, and depicts no existing place.

Three things the panel must answer

1. What it is doing
2. How the data is handled
3. Who supervises it

Where the three signage principles land

Signage for AI installations cannot simply state a name. The explainability-first principle becomes a concrete requirement here: the panel must answer the three questions above. The observation point itself collects no visitor data — open to watch is not open data. Entry is ramped with no steps, on the same accessibility baseline as the chevron ramp.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

The sign must answer three questions

What is it doing; how is data handled; who supervises it. A device missing any of the three does not enter a public-space pilot. It is the only hard requirement in the wayfinding system.

What is not collected

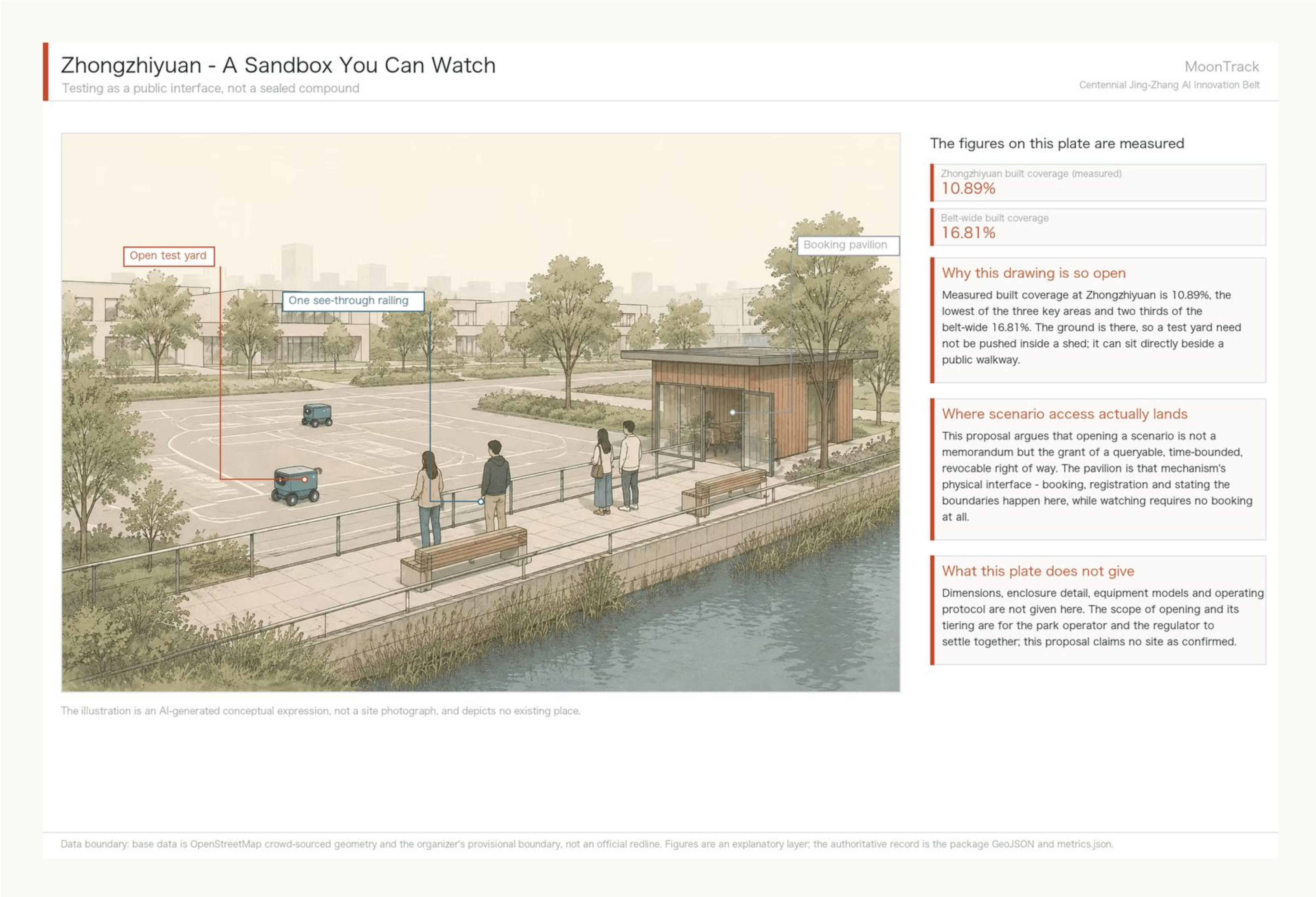
No faces, no individual movement traces. An observation point is where people watch robots, not where robots watch people: that directional distinction sets its orientation, seating and sightlines.

Why it needs a place of its own

Public acceptance is a risk shared by all ten scenario cards. Putting the robots somewhere people can stop, ask and object works better than explaining in a brochure. The illustration is an AI-generated concept image, not a photograph.

Zhongzhiyuan: A Sandbox You Can Watch

A built coverage of 10.89% is what allows this arrangement



Watching needs no booking; entering does

The test yard is separated from the public walkway by one see-through railing. Opening a scenario is not a memorandum but the grant of a queryable, time-bounded, revocable right of way, and the pavilion is that mechanism's physical interface.

Why this drawing is so open

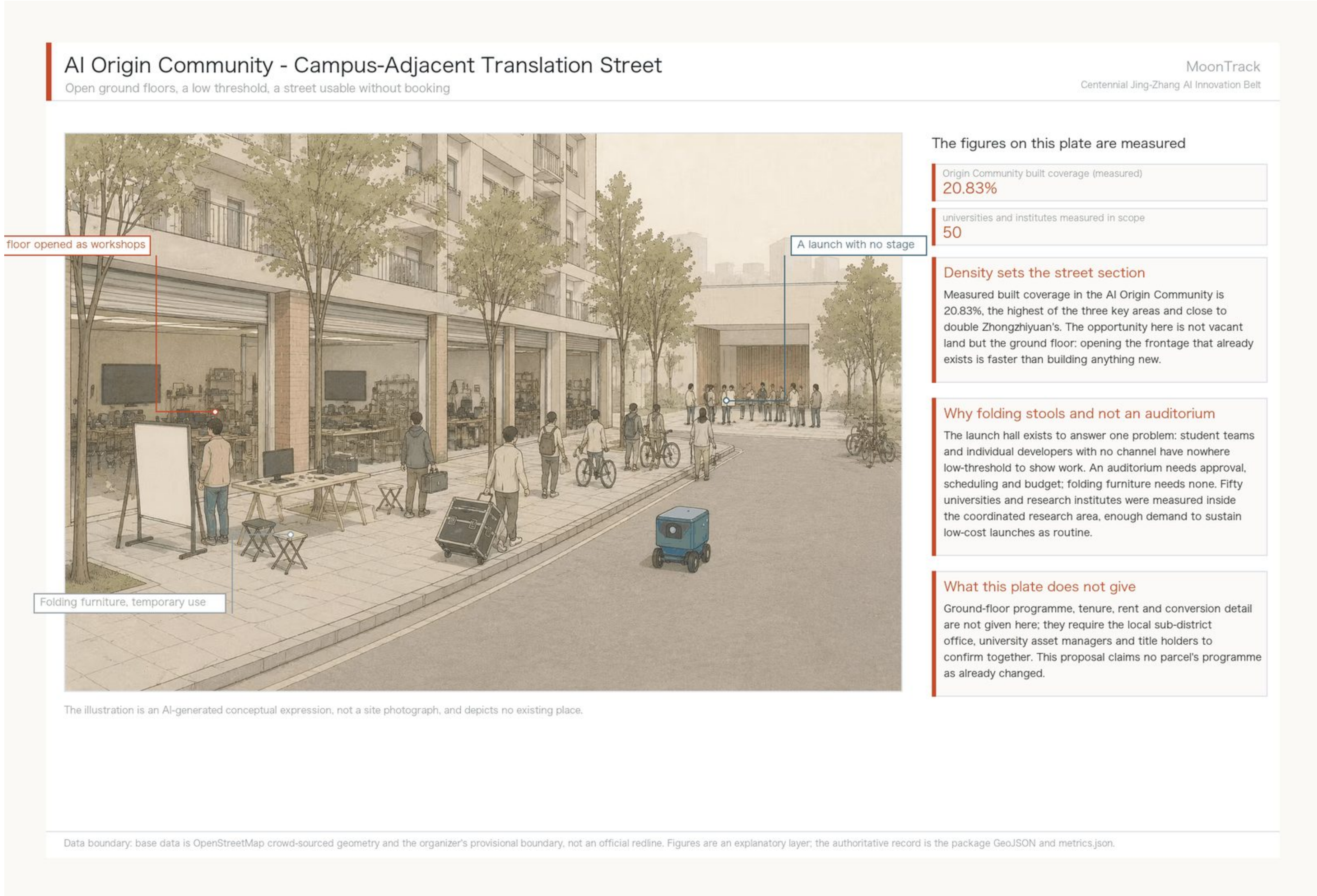
Measured built coverage is 10.89%, the lowest of the three key areas and two thirds of the belt-wide 16.81%. The ground is there, so a test yard need not be pushed inside a shed.

What this sheet does not give

Dimensions, enclosure, equipment models and operating protocol are not given. Scope and tiering of opening are for the operator and regulator to settle. The illustration is an AI-generated concept image, not a photograph.

AI Origin Community: Translation Street

At 20.83%, the opportunity is the ground floor, not vacant land



Open the frontage that already exists

Measured built coverage is 20.83%, the highest of the three and close to double Zhongzhiyuan's. Opening the ground floor is faster than building anything new.

Why folding stools and not an auditorium

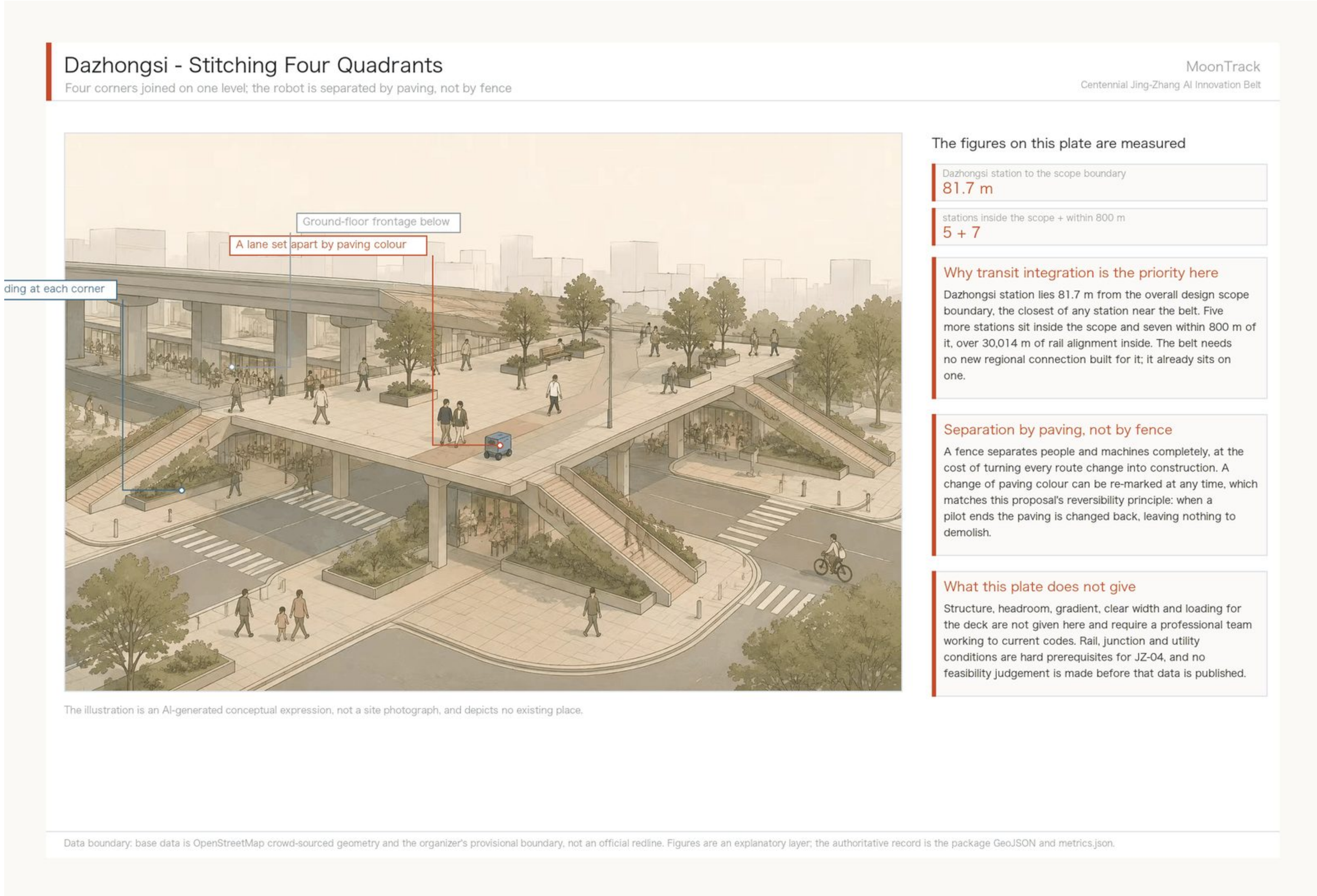
The problem is that student teams and individual developers with no channel have nowhere low-threshold to show work. An auditorium needs approval, scheduling and budget; folding furniture needs none. Fifty universities and institutes were measured inside the coordinated research area.

What this sheet does not give

Programme, tenure, rent and conversion detail are not given; they need the sub-district office, university asset managers and title holders to confirm. The illustration is an AI-generated concept image, not a photograph.

Dazhongsi: Stitching Four Quadrants

The station is 81.7 m out, so the move is joining one level



Separation by paving, not by fence

A fence separates people and machines completely, at the cost of turning every route change into construction. Paving can be re-marked at any time; when a pilot ends it is changed back, leaving nothing to demolish.

Why transit integration is the priority here

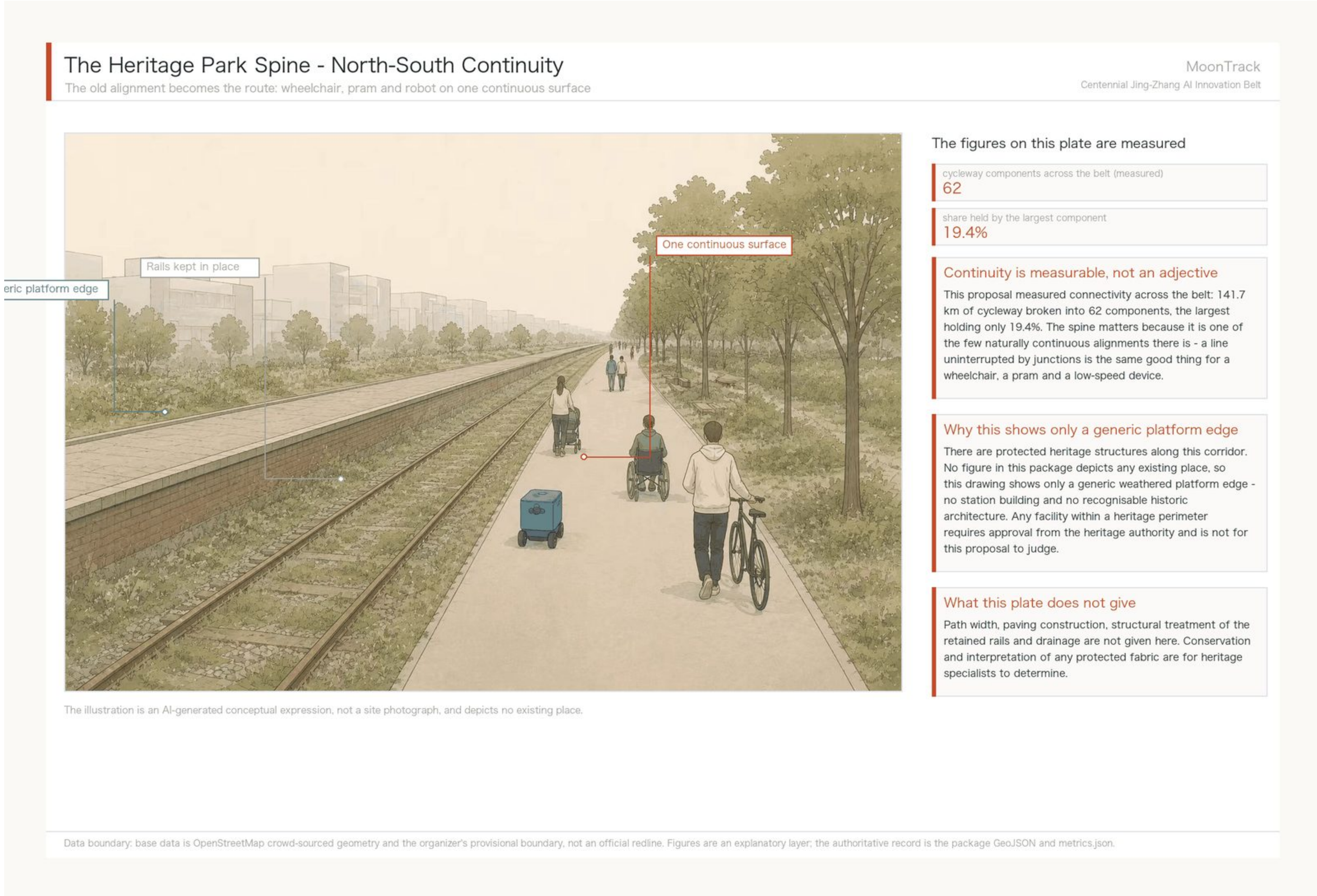
Dazhongsi station lies 81.7 m from the scope boundary, the closest of any station near the belt. Five more sit inside the scope and seven within 800 m, over 30,014 m of rail alignment inside.

What this sheet does not give

Structure, headroom, gradient, clear width and loading are not given. Rail, junction and utility conditions are hard prerequisites for JZ-04, and no feasibility judgement is made before that data is published. The illustration is an AI-generated concept image, not a photograph.

The Heritage Park Spine: Continuity

The old alignment becomes the route: wheelchair, pram and robot on one surface



Continuity is measurable

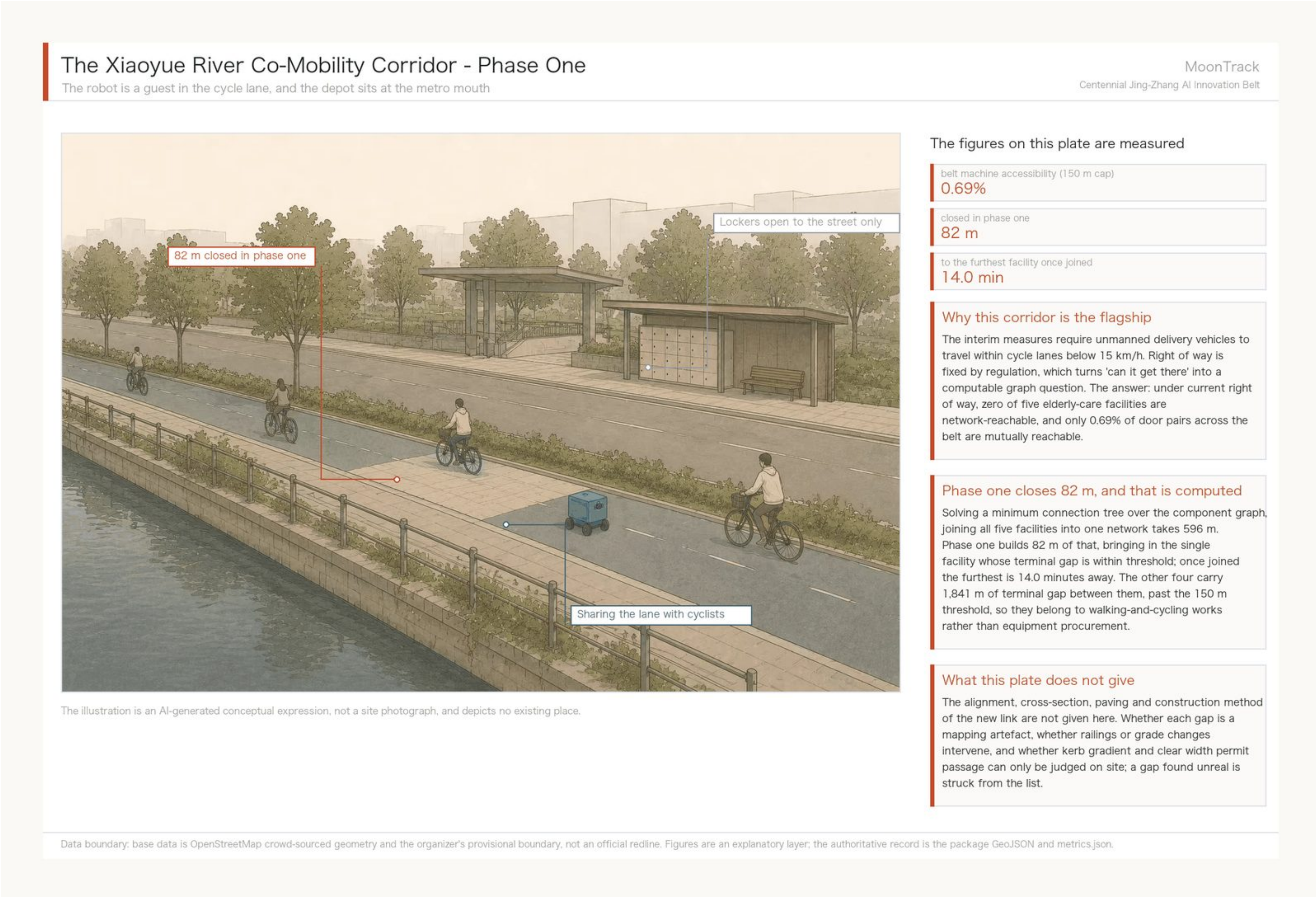
Across the belt, 141.7 km of cycleway breaks into 62 components with the largest holding only 19.4%. The spine matters because it is one of the few naturally continuous alignments there is.

Why only a generic platform edge

Protected heritage structures do stand along this corridor. No figure in this package depicts any existing place, so only a generic weathered platform edge appears and no station building. Facilities within a heritage perimeter require heritage-authority approval.

What this sheet does not give

Path width, paving, structural treatment of the retained rails and drainage are not given. Conservation and interpretation of protected fabric are for heritage specialists. The illustration is an AI-generated concept image, not a photograph.



Phase one closes 82 m, and that is computed

Joining all five facilities into one network takes 596 m. Phase one builds 82 m of it, bringing in the single facility whose terminal gap is within threshold; once joined the furthest is 14.0 minutes away. The other four carry 1,841 m between them, past the 150 m threshold, so they belong to walking-and-cycling works.

Why this corridor is the flagship

Right of way is fixed by regulation, which turns whether it can get there into a computable graph question. The answer: zero of five elderly-care facilities are network-reachable today, and only 0.69% of door pairs across the belt are mutually reachable.

What this sheet does not give

Alignment, cross-section, paving and construction method of the new link are not given. Whether each gap is a mapping artefact can only be judged on site, and one found unreal is struck from the list. The illustration is an AI-generated concept image, not a photograph.

Renewal Project List and Phasing

JZ-07 is the smallest and earliest item

ID	Project	Phase	Type
JZ-07	**Field verification of three key network gaps**	**Near term**	**Mobility**
JZ-01	Heritage-park slow-mobility stitching	Near term	Public space
JZ-02	Zhongzhiyuan Qinghe innovation interface	Medium term	Blue-green
JZ-03	Campus-adjacent translation street	Medium term	Renewal
JZ-04	Dazhongsi four-quadrant pedestrian links	Medium term	Transit
JZ-05	AI service and edge-compute nodes	Long term	Infrastructure
JZ-06	Global AI week public route	Ongoing	Operations

Why JZ-07 comes first

Measurement shows that 56 m of connection across three points could grow the usable robot network from 27,560 m to 57,770 m, a return no other item on the list approaches. It must nevertheless begin with field verification: whether the breaks are mapping artefacts, whether railings or grade changes intervene, and whether kerb gradient and clear width permit passage can only be judged on site. It proceeds only if verification confirms the breaks, and is struck from the list otherwise. Almost all of its cost is verification, not construction.